fMRI BOLD signals classification in visual stimuli categories

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***Abstract*—In the neuroscientific field a central question is about how information is represented in the brain. One investigation method is fMRI. In particular, EfMRI (Event-related fMRI) is able to give information about the functionality of the brain, by measuring the changes in the deoxyhemoglobin concentration consequent to task-induced or spontaneous modulation of neural metabolism, called BOLD signal.**
In the BOLD5000 study, almost 5000 images belonging to three different datasets have been presented to patients in different sessions, while fMRI was registered on them.
The aim of this paper is to propose different methods for applying the following pipeline: i) uniform the labels of the three used datasets, ii) extract a measure of the responses from the BOLD volumes, iii) implement classification models to predict the label of the visual stimulus, analyzing the neural responses of the patient to that image.

1. Introduction

This work follows the pipeline proposed by Qiao et al. [3]. In the human visual system, when people process visual content, information flows from primary visual cortices to higher-level visual cortices and vice versa, following bottom-up and top-down processing, respectively. Inspired by these bidirectional information flows, Qiao et al. [3] proposed a bidirectional recurrent neural network (BRNN)-based method to decode the corresponding visual categories from fMRI data.
This work mimics that pipeline, applying it to a different dataset than the one used in [3]; this introduces additional challenges in the labeling procedure that had to be addressed. Furthermore, the approach is extended beyond the BRNN model with a Graph Neural Network (GNN)-based method.
This project was developed individually as part of a university exam, and therefore presents some limitations.

2. BOLD5000: A public fMRI dataset of 5000 images

The BOLD5000 dataset is a large-scale, slow event-related human fMRI study that incorporates 5000 real-world images as stimuli. [2]

2.1. Visual Stimuli

The visual stimuli presented have been taken from three pre-existing computer vision datasets: COCO, ImageNet and SUN.

- *Scenes*. 1000 images of scenes have been taken from the SUN datasets. They include indoor and outdoor environments (e.g. restaurants, mountains).
- *COCO*. 2000 images from COCO have been used, which are multi-labels images, meaning that they contain multiple boxes around multiple objects.
- *ImageNet*. 1916 images have been taken from ImageNet, which represent one single subject in the image.

The images have undergone selection and pre-processing steps accurately described in [2], before being showed to the patients.

2.1.1. Compatibility issues among the three datasets

The integration of COCO, ImageNet, and SUN poses several challenges due to their intrinsic differences.
COCO is a multi-label dataset, where each image may contain multiple objects annotated and the respective bounding boxes. This substantially changes the nature of the classification task compared to single-label datasets.
ImageNet, on the other hand, provides highly specific categories (958 classes), such as distinct species of fish rather than the general category "fish." This level of granularity often reduces the effectiveness of language-based encoding models, which struggle to capture fine-grained categorizations.
Finally, SUN consists of scenic images, showing indoor and outdoor environments rather than individual objects. The fundamental difference of these stimuli from the other datasets complicates the integration of labels across the three sources. An example is shown in Figure 1.

2.2. Experimental Design

In [2], fMRI data were collected from four participants across multiple scanning sessions. Each session consisted

Figure 1: From the top to the bottom: examples of images from COCO, ImageNet, SUN, with the respective labels.

of several runs: task-related runs, designed to acquire data during the presentation of the visual stimuli described above, and localizer runs, which employed different stimuli to define specific regions of interest in the brain.
Figure 2 shows the structure of a slow event-related task.

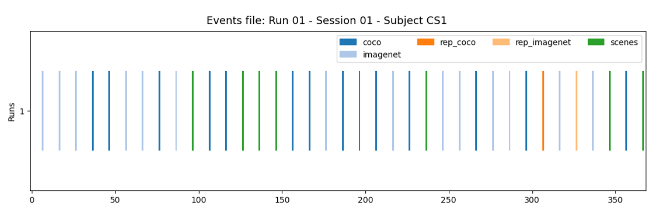

Figure 2: Structure of slow event-related task. The colors differentiate the visual stimuli showed in each time point based on the original dataset.

For each hemisphere, five regions of interest (ROIs) were identified, resulting in a total of ten ROIs per participant. ROIs are functionally defined cortical areas that exhibit selective responses to specific categories of visual stimuli. In fMRI research, extracting the BOLD signal from ROIs allows for targeted analyses focused on brain regions known to be relevant for the experimental task, increasing sensitivity compared to whole-brain voxel-wise approaches.

In particular:

- *Scene-selective regions* PPA, RSC, OPA were defined using the contrast of scenes compared with objects and scrambles.
- *Object-selective region* LOC was defined by comparing objects to scrambled images.
- *Early visual region* EarlyVis was defined by comparing the scrambled images to a baseline.

During the task-related runs, each image was presented for 1 second, followed by a 9-second fixation cross before the next stimulus. This slow event-related design allowed extraction of the time course of the BOLD signal, revealing the hemodynamic response's peak about 6 seconds after stimulus onset, and the consequent return to baseline.

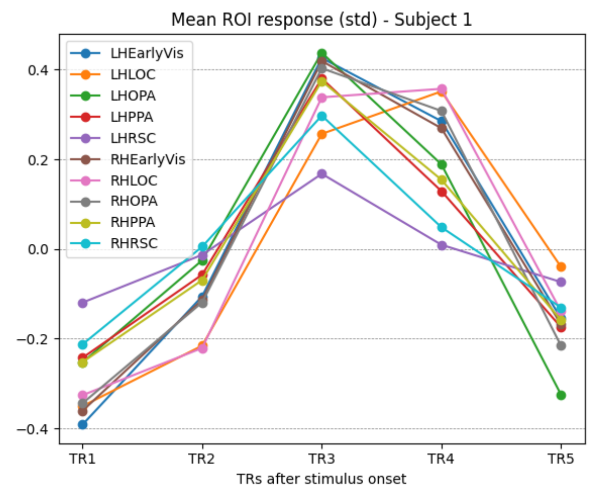

Figure 3: Mean standardized BOLD signal across trials for each ROI in Subject 1, plotted over five consecutive TRs following stimulus onset. The signal was averaged across trials and standardized (z-score) to highlight relative changes in activation. Most regions show a peak at TR3, consistent with the expected hemodynamic response.

3. Dataset Preparation

3.1. Solving the Labels Problem

Four target categories were defined a priori onto which to map the labels of all three datasets: *object*, *animal or human*, *plant or food*, *scene*.
Since the original dataset labels consist of single words or short phrases rather than full sentences, directly embedding them with a sentence-level model would have been suboptimal. To address this, each label was converted into a short descriptive text. For instance, instead of embedding the ImageNet class *tinca tinca* directly, the text *"a photo of a tinca tinca"* was used. This format was applied uniformly to all labels across all three datasets. Similarly, prototype texts were created for each of the four final categories. For example, *"animal or human"* was represented by texts such as *"a photo of a pet"* and *"a photo of a person"*.

These embeddings were computed using *all-MiniLM-L6-v2* [4], a sentence transformer model that maps sentences and paragraphs to a 384-dimensional dense vector space, suitable for tasks such as semantic search and clustering. The model has 22.7M parameters. Each label text was then assigned to the final category whose prototype embedding was closest, measured by cosine similarity.

3.1.1. ImageNet

For ImageNet, I followed a single two-step procedure to map the original 958 fine-grained categories into the four final groups. First, I used WordNet (via NLTK) to generalize each ImageNet class. Since every ImageNet label is linked to a WordNet ID, I retrieved the corresponding synset and moved up the hypernym hierarchy to obtain a more abstract category.

Level	Synset Hierarchy
0	entity.n.01
1	abstraction.n.06
2	relation.n.01
3	part.n.01
4	substance.n.01
5	material.n.01
6	paper.n.01
7	tissue.n.02
8	toilet_tissue.n.01

TABLE 1: Example of WordNet hypernym hierarchy for the ImageNet class *toilet_tissue*. A *hypernym* is a more general concept (e.g., *paper* is a hypernym of *toilet tissue*), while a *hyponym* is a more specific instance.

By testing different levels of generalization, I selected the level that produced a reasonable number of coherent super-categories. This choice was quite arbitrary, in the measure that I tried different levels, analyzed some mappings, and decided that level 6 was the most reasonable. At level 2, the unique categories created were 6, till from level 16 they were all 958 categories, same as original ImageNet dataset labels. In the second step, these WordNet-based super-categories have been mapped to the four final labels using a sentence-embedding model (SentenceTransformer all-MiniLM-L6-v2). I created short descriptive texts for each final category, computed prototype embeddings, and compared them with the embeddings of the ImageNet super-categories. Each class was then assigned to the closest final label based on cosine similarity. This combined procedure allowed me to reliably integrate ImageNet into the unified labeling scheme.

3.1.2. COCO

The COCO dataset is more difficult to integrate because its images are multi-label: each picture often contains several objects at the same time. I first tried to map COCO categories to the four final groups using sentence-embedding similarity, but the results were not reliable when using single words. To improve the mapping, short descriptive texts for each final category have been defined and prototype embeddings obtained by averaging their representations. This approach worked better, and most COCO labels were correctly assigned to one of the four groups. The remaining outliers could be handled by lowering the similarity threshold. However, the main issue remains that COCO images contain multiple objects, while my classification task requires a single label per stimulus. For this reason, I decided not to use fMRI data collected from COCO stimuli and to focus only on ImageNet and SUN images.

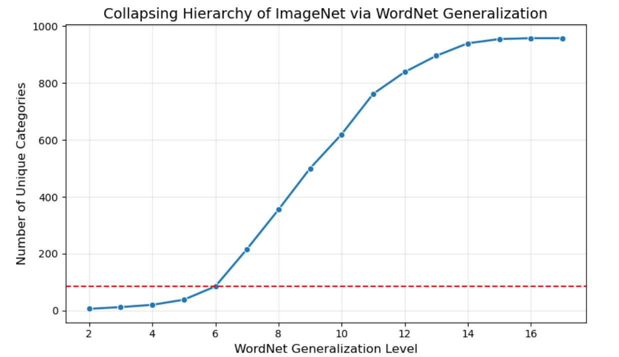

Figure 4: Number of ImageNet categories obtained at different WordNet generalization levels. The red dashed line marks the selected level.

3.1.3. SUN

The SUN dataset contains only scene categories, such as indoor and outdoor environments, and does not include objects or animals like the other datasets. I first tried to map SUN labels to the four final categories using sentence-embedding similarity, but this approach did not work because the SUN labels are semantically too different from the other groups. I then explored the structure of the SUN labels by embedding them and clustering them into semantic topics, but the clusters still represented only different types of scenes. For this reason, I decided to assign all SUN categories to a single final label, “scene”, which is consistent with the nature of the dataset and suitable for the classification task.

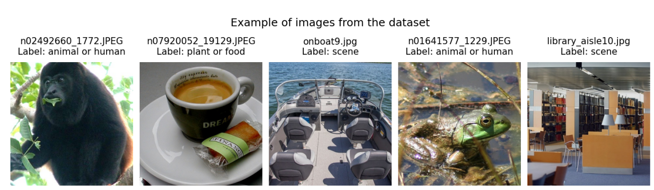

Figure 5: Examples of five random images with their corresponding names and final label.

3.2. Final Labels Distribution

After discarding the COCO dataset, all the labels of ImageNet and SUN datasets were mapped into one of the four final categories chosen. Figure 6 shows the mapping distribution.

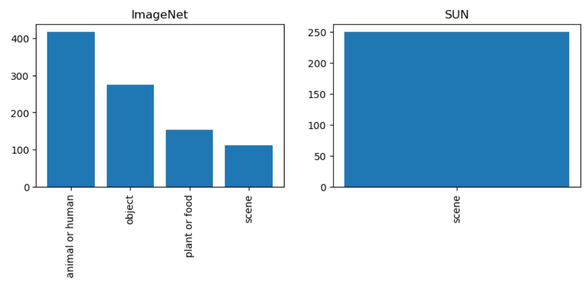

Figure 6: Distribution of mapped labels for SUN and ImageNet. The left subplot shows how ImageNet categories were assigned to each of the four final labels using the prototype-based similarity approach. The right subplot reflects the decision to map each SUN category into the final label 'scene'.

4. Data Exploration

The fMRI dataset consists of 4D functional images, where the first three dimensions represent the brain volume and the fourth dimension represents time. Each run is therefore a sequence of 3D volumes acquired at regular intervals (one volume per TR), forming a time series of BOLD signal changes across the whole brain. In addition to the functional data, the dataset also includes a set of region-of-interest (ROI) masks that identify specific cortical areas relevant for visual processing.

4.1. Voxel Responses Extraction

Since the goal of the project is to classify the visual stimuli shown to the participants, the 4D fMRI signals needed to be converted into one feature vector per trial. To achieve this, I applied a Least-Squares-Separate (LSS) General Linear Model (GLM) to each run. For every single trial, I built a small GLM with one regressor for the target trial and another regressor grouping all the remaining trials. This approach allowed me to estimate a beta map for each trial, representing how strongly each voxel responded to that specific stimulus.
The GLM models the BOLD signal as a combination of:

- regressors representing the presented stimuli,
- regressors accounting for confounds,
- a hemodynamic response function (HRF) describing how the brain responds over time.

$$\mathrm{BOLD}(t) = X\beta + \epsilon$$

where X is the design matrix (stimuli + confounds), β are the estimated weights (beta values), and ϵ is noise. The GLM estimates β, which indicates how much each regressor contributes to the signal in each voxel.
Once the beta map was computed, I applied the ROI masks using a NiftiMasker. Each ROI selects a specific brain region, and the masker extracts only the voxels inside that region. The voxel values from all ROIs were then flattened and concatenated into a single feature vector. In this way, every trial is represented by a long vector containing the activation pattern across all selected brain areas.

4.2. MLP

A multi-layer perceptron (MLP) is a type of feedforward neural network composed of fully connected layers of neurons. It begins with an input layer, whose size is determined by the dimensionality of the input data.
In this case, the input data consisted of feature vectors extracted from fMRI voxel responses. These vectors were created by concatenating the voxel values from multiple ROI masks. Because each ROI contains a different number of voxels for each subject, the resulting feature vectors varied in length across subjects. They were consistent only across trials within the same subject.
To make the input dimensionality compatible with a fixed-size MLP architecture, I applied Principal Component Analysis (PCA) and projected each feature vector into a 1024-dimensional space. This ensured that all subjects produced feature vectors of identical length, matching the required input size of the MLP.
The MLP then processes these vectors through one or more hidden layers, where each neuron receives input from all neurons in the previous layer. These hidden layers apply nonlinear transformations, enabling the network to learn complex relationships in the data.
Finally, the output layer contains a number of neurons corresponding to the classification task, in this case four neurons.

4.2.1. Grid Search

Hyperparameter	**Values Tested**
Hidden layer sizes	(512, 256, 128, 64), (256, 128, 64), (256, 64), (128, 64)
Activation function	relu, logistic, tanh
Solver	sgd, adam
Regularization α	0.00001, 0.0001, 0.001, 0.01
Batch size	32, 64, 128
Total combinations	**288**

TABLE 2: Hyperparameter grid used for the MLP grid search.

The best parameters have been chosen based on the F1 score on the validation set, which is a metric used to evaluate classification models, in particular when there are imbalanced classes.

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

The grid search identified the following configuration as optimal: activation = logistic, $\alpha = 0.01$, batch_size = 32, hidden_layer_sizes = [256, 64], solver = Adam.

4.2.2. Results

The model was evaluated on the test set, obtaining an accuracy of 0.44, a precision of 0.26, a recall of 0.25, and an F1-score of 0.25. These results indicate that the model is unable to perform well on the classification task. The slightly higher accuracy is likely due to class imbalance, as confirmed by the low F1-score, which is equal to random guessing.

5. Hierarchical approach

Inspired by [3], the goal of this component is to model the bidirectional (bottom–up and top–down) information flow that characterizes hierarchical processing in the human visual system.

Unlike the MLP model, where input feature vectors were obtained by concatenating voxel responses across all ROIs, here the emphasis is placed on the *sequential* structure of the data. Instead of collapsing all regions into a single vector, the input is divided into distinct ROI-specific segments, preserving the anatomical and functional hierarchy of the visual cortex.

To capture this structured organization, I implemented:

- a Bidirectional Recurrent Neural Network (BRNN), to explicitly model bottom–up and top–down information flow;
- a multi-branch GNN, allowing parallel processing of ROIs of the two different hemispheres before integration;

5.1. Visual Encoding

The visual encoding step serves as a bridge between the image stimuli and the voxel responses. The goal is to obtain a compact, semantically meaningful representation of each stimulus image, which can then be used to identify the most informative voxels for each ROI. Specifically, for each voxel, an encoding model is fitted to predict its activation from the image features. Voxels whose activations are better predicted by the image features are considered more informative for the visual decoding task, and are therefore selected as input to the decoding models. This approach follows the methodology proposed by [3].

5.1.1. Visual Encoding: Custom CNN

A lightweight custom CNN was designed to classify the stimulus images into the four target categories. The architecture consists of two convolutional blocks (`Conv2d` → `BatchNorm` → `ReLU` → `MaxPool2d`), followed by an `AdaptiveAvgPool2d` layer that reduces the spatial dimensions to a fixed 4×4 grid, and two fully connected layers producing a 32-dimensional embedding before the final classification head. Dropout ($p = 0.5$) was applied for regularization.

A grid search over 16 hyperparameter configurations was performed, selecting the best model based on macro F1-score on the validation set. The best configuration used lr $=$ 0.001, Adam optimizer, num_filters $= 32$, and batch size $= 64$. The final model was trained for 100 epochs with weighted cross-entropy loss to address class imbalance. Test accuracy: 0.954, macro F1-score: 0.948.

5.1.2. Visual Encoding: Pre-trained ResNet-50

As an alternative visual encoder, a ResNet-50 pre-trained on ImageNet was used. All convolutional layers were frozen, and only the final fully connected layer was replaced with a new `Linear(2048 → num_classes)` layer and fine-tuned on the BOLD5000 stimulus set for 40 epochs, using the same hyperparameters identified during the CNN grid search.

For feature extraction, the output of the average pooling layer — immediately before the classification head — was used, yielding a 2048-dimensional vector per image. Since all convolutional layers were frozen throughout training, these features correspond to the original ImageNet pre-trained representations, unmodified by the fine-tuning step. Images were preprocessed using standard ImageNet normalization (resize to 256, center crop to 224). Test accuracy: 0.950, macro F1-score: 0.942.

5.1.3. ResNet-50 vs Custom CNN

	Fine-Tuned ResNet-50	**Custom CNN**
Pretraining	ImageNet	None
Trainable params	Final FC layer only	Entire network
Feature dim.	2048-D	32-D
Epochs	40	100
Test Accuracy	0.950	0.954
Test F1 Score	0.942	0.948

TABLE 3: Comparison of the two visual encoding strategies.

The two models achieved comparable performance. ResNet-50 was selected as the feature extractor for the subsequent decoding pipeline for two reasons: it produces higher-dimensional representations (2048-D vs 32-D), which are likely to be more semantically rich. Using pretrained ImageNet features also avoids introducing additional sources of noise that could propagate to the fMRI decoding stage.

5.1.4. ROI-based Voxel Selection

Because the decoding models require fixed-size inputs, a voxel selection procedure was applied to identify the most informative voxels within each ROI or ROI pair. Each ROI pair contains the left and right hemisphere counterparts of the same visual area.

For each subject, ResNet-50 image features were used to fit a multi-output Ridge Regression model predicting voxel responses from image features. The regularization parameter α was chosen via `RidgeCV` on a random subset of 500 training samples, maximizing the average R^2 across folds. Voxels were then ranked by their Pearson correlation between predicted and true responses on the validation set, and the top-k were retained. The procedure was applied independently per ROI or ROI pair.

This procedure was applied differently for the two decoding architectures:

- **BRNN**: selection applied per ROI *pair*; top 100 voxels per pair, yielding fixed-length input vectors of dimension 100 at each time step.
- **Multibranch GNN**: selection applied per single ROI; top 51 voxels per ROI, so that each hemisphere branch processes a graph with 5 nodes of consistent feature size.

The sequence of ROI pairs follows the well-established hierarchical organization of the human ventral visual stream. Early visual areas (V1–V3) constitute the lowest level of the hierarchy, encoding local visual features such as edges and orientations. The Lateral Occipital Complex (LOC) represents a higher-level stage involved in object and shape processing. The Occipital Place Area (OPA) encodes navigationally relevant scene structure. The Parahippocampal Place Area (PPA) processes global scene layout and contextual information. Finally, the Retrosplenial Cortex (RSC) integrates allocentric spatial representations and memory. This progression from EarlyVis $\rightarrow$ LOC $\rightarrow$ OPA $\rightarrow$ PPA $\rightarrow$ RSC reflects the canonical feedforward hierarchy of visual processing in the human brain [1].

$$\begin{tikzcd}
\begin{array}{c}\text{LHEarlyVis}\\ \text{RHEarlyVis}\end{array} \arrow[r] & \begin{array}{c}\text{LHLOC}\\ \text{RHLOC}\end{array} \arrow[r] & \begin{array}{c}\text{LHOPA}\\ \text{RHOPA}\end{array} \arrow[d] \\
& \begin{array}{c}\text{LHRSC}\\ \text{RHRSC}\end{array} & \begin{array}{c}\text{LHPPA}\\ \text{RHPPA}\end{array} \arrow[l]
\end{tikzcd}$$

Figure 7: Sequence of ROI pairs used as BRNN time steps, following the hierarchical order of the ventral visual stream.

6. Visual Decoding

Three decoding architectures were evaluated: a Bidirectional Recurrent Neural Network (BRNN), a Graph Neural Network (GNN), and a Multibranch GNN with two separate branches, one per hemisphere, using single ROIs rather than ROI pairs. All models were trained and evaluated on the same train/val/test split. Class imbalance was addressed using weighted cross-entropy loss, with class weights computed from the training set distribution.

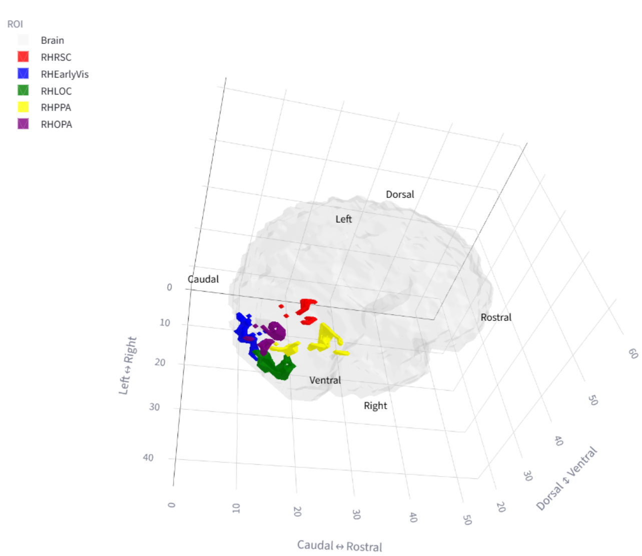

Figure 8: Visualization of ROIs from a dashboard.

6.1. BRNN

The BRNN architecture was adopted following the approach proposed by [3], which demonstrated the effectiveness of bidirectional recurrent networks for decoding visual category information from fMRI BOLD signals. The selected voxels of each ROI pair form the input sequence, allowing the model to exploit both the hierarchical organization of the visual system and the bidirectional information flows — bottom-up and top-down — characteristic of visual cortices.

6.1.1. Model Architecture

Each trial is encoded as a sequence of 5 ROI pairs, each providing a fixed-size vector of 100 selected voxels, yielding an input of shape (sequence length $= 5$, feature size $= 100$). The architecture consists of a multi-layer bidirectional LSTM, with dropout applied within the recurrent layers. The concatenated forward and backward hidden states at the last time step are passed to a fully connected classification layer:

$$y = \mathrm{FC}\big([\overrightarrow{h}_T \parallel \overleftarrow{h}_1]\big), \quad \text{where } \overrightarrow{h}_T, \overleftarrow{h}_1 \in \mathbb{R}^h,$$

with h denoting the hidden size, so that the input to the classifier has dimension $2h$.
Formally, given an input sequence

$$X = \{x_1, x_2, \ldots, x_T\}, \quad x_t \in \mathbb{R}^{100},$$

the bidirectional LSTM processes the sequence in both directions:

- **forward**: EarlyVis $\rightarrow$ LOC $\rightarrow$ OPA $\rightarrow$ PPA $\rightarrow$ RSC
- **backward**: RSC $\rightarrow$ PPA $\rightarrow$ OPA $\rightarrow$ LOC $\rightarrow$ EarlyVis

Each time step thus has access to context from the entire visual hierarchy, both from lower and higher areas simultaneously.

6.1.2. Hyperparameter Search

A grid search over 64 configurations was performed. The best model was selected based on macro F1-score on the validation set.

Hyperparameter	Values Tested
Normalization	z-score, min-max
Hidden size	16, 32
Number of layers	1, 2
Dropout	0.5
Batch size	32, 64
Optimizer	Adam, SGD
Learning rate	0.001, 0.0001
Total combinations	**64**

TABLE 4: Hyperparameter grid for the BRNN model search.

Best configuration: batch size $= 32$, hidden size $= 32$, learning rate $= 0.001$, normalization $=$ z-score, number of layers $= 1$, optimizer $=$ Adam.

6.1.3. Training Procedure

The model was trained for 100 epochs using weighted cross-entropy loss to compensate for class imbalance. Optimization was performed with the Adam optimizer, and the learning rate was adaptively reduced using a `ReduceLROnPlateau` scheduler (factor 0.8, patience 5).

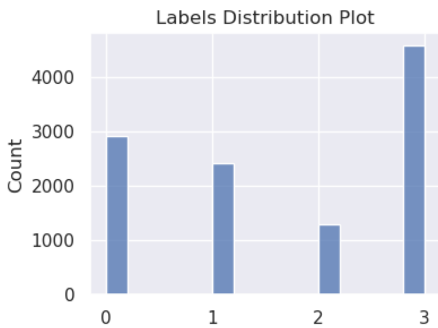

Figure 9: Distribution of the dataset used in all decoding models.

6.1.4. Results

Metric	Test
Accuracy	0.34
Macro F1-score	0.28
Precision	0.30
Recall	0.29

TABLE 5: BRNN test performance.

The BRNN achieved results only marginally above chance level. Training metrics improved over epochs, but validation performance did not follow, suggesting a generalization problem likely attributable to the limited dataset size. The rationale behind the architecture remains solid: the sequential structure of the ROI pairs is well-suited to a bidirectional model, which can capture both bottom-up and top-down interactions across the visual hierarchy.

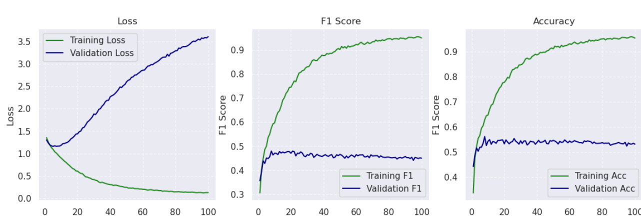

Figure 10: BRNN training curves.

6.2. Multibranch Graph Neural Network

While the BRNN treats ROIs as an ordered sequence, assuming a fixed hierarchical structure, an alternative approach is to model brain regions as a network of interconnected areas, without imposing a predefined processing order. A Graph Neural Network (GNN) naturally supports this formulation, representing ROIs as nodes and their functional or structural relationships as edges.

A standard GNN, however, merges all ROIs into a single graph, combining left and right hemisphere regions without distinction. To address this, a Multibranch GNN was developed, introducing two parallel branches — one per hemisphere — that process left and right ROIs independently before combining their representations.

6.2.1. Model Architecture

The model consists of two parallel Graph Attention Network (GAT) branches, one for the left hemisphere and one for the right. A GAT is a type of GNN that uses an attention mechanism to learn how much each neighboring node should influence a node's updated representation. Rather than treating all adjacent nodes as equally informative, a GAT computes attention coefficients that quantify the importance of each connection, making it particularly suitable for brain connectivity data where not all region pairs contribute equally.

Each branch receives as input a graph with 5 nodes (one per ROI) and node features of dimension 51 (the top-51 selected voxels per ROI), and applies one or two GAT convolution layers, each followed by BatchNorm, ReLU, and Dropout. A global mean pooling step then aggregates the node embeddings into a single hemisphere-level embedding of dimension 32. The two hemisphere embeddings are concatenated and passed to a fully connected classifier:

$$y = \mathrm{FC}\big([\mathbf{e}_{\mathrm{LH}} \parallel \mathbf{e}_{\mathrm{RH}}]\big), \quad \mathbf{e}_{\mathrm{LH}}, \mathbf{e}_{\mathrm{RH}} \in \mathbb{R}^{32}.$$

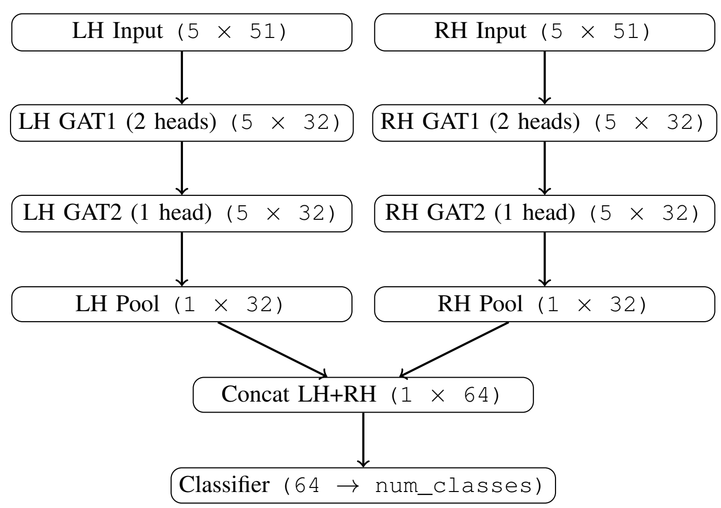

Figure 11: Architecture of the Multibranch GNN. Each hemisphere is processed by an independent GAT-based branch; the resulting embeddings are concatenated and passed to a linear classifier.

6.2.2. Hyperparameter Search

A grid search over 64 configurations was performed. The best model was selected based on macro F1-score on the validation set.

Hyperparameter	Values Tested
Normalization	z-score, min-max
Hidden channels	8, 16
Number of layers	1, 2
Dropout	0.5
Batch size	32, 64
Optimizer	Adam, SGD
Learning rate	0.001, 0.0001
Total combinations	**64**

TABLE 6: Hyperparameter grid for the Multibranch GNN model search.

Best configuration: batch size = 32, hidden channels = 16, learning rate = 0.001, normalization = z-score, number of layers = 2, optimizer = Adam.

6.2.3. Training Procedure

Three variants of the model were trained using the best hyperparameters found during grid search, for 150 epochs each, varying the loss function and the use of graph augmentation:

- weighted cross-entropy loss, no augmentation;
- weighted cross-entropy loss, with augmentation on-the-fly;
- focal loss ($\gamma = 8.0$), with augmentation on-the-fly.

Graph augmentation consisted of edge dropout ($p = 0.4$), feature masking ($p = 0.3$), and Gaussian noise (std = 0.02), applied on-the-fly during training, as shown in Figure 12.

6.2.4. Results

Loss	Aug	Acc	F1	Prec	Rec
CE	No	0.49	0.39	0.42	0.43
CE	Yes	0.51	0.35	0.36	0.40
Focal ($\gamma = 8.0$)	Yes	0.54	0.32	0.27	0.40

TABLE 7: Multibranch GNN test performance across training configurations.

The baseline model trained with cross-entropy loss and no augmentation achieved the best macro F1-score (0.39), despite not having the highest accuracy. All three variants struggle to correctly classify the least frequent classes, suggesting that the difficulty is not fully resolved by augmentation or focal loss. The augmentation may be too aggressive, reducing the discriminative differences between class signals. The overall limited performance is consistent with the challenges discussed in Section 7.1: small dataset size, label noise introduced by the automatic label mapping procedure, and the inherent difficulty of decoding fine-grained semantic categories from a limited number of fMRI trials.

7. Conclusion

7.1. Challenges and Future Directions

Voxel Extraction Strategy. The voxel extraction pipeline required computing a beta map for each individual trial via the LSS procedure, which was computationally expensive. Faster alternatives exist, such as using a single GLM across

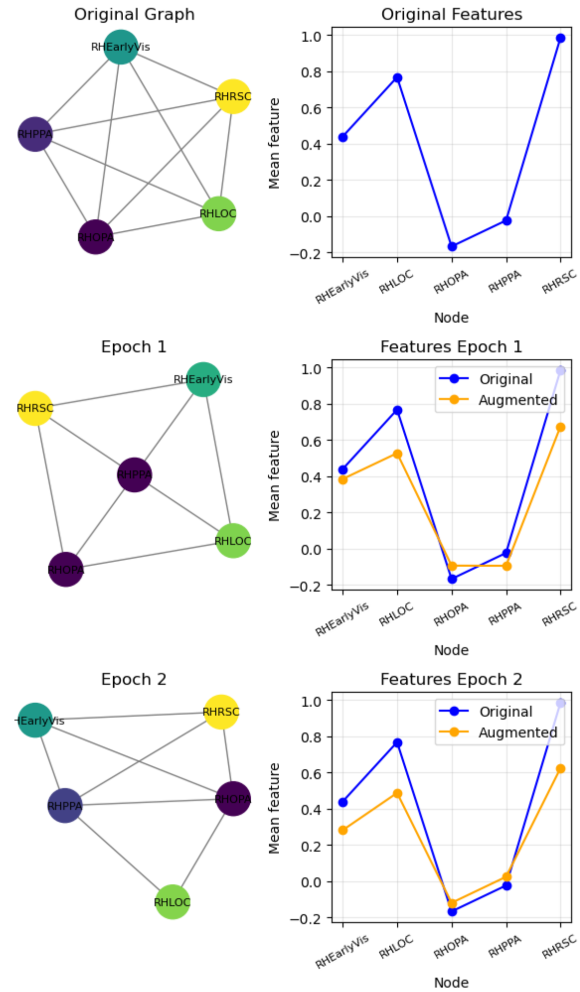

Figure 12: Augmentation during Multibranch GNN training.

all trials, but they would have sacrificed trial-level specificity. Given that this representation is the foundation of all subsequent decoding steps, the more expensive approach was retained to preserve as much information as possible. This is primarily a *neuroscientific* limitation.

Assumption on ROI Ordering. The hierarchical sequence EarlyVis → LOC → OPA → PPA → RSC was assumed to reflect a meaningful processing order. However, this ordering was implicitly provided by the dataset structure and may not fully capture the actual organization of visual processing in individual subjects. The introduction of the Multibranch GNN was partly motivated by the desire to reduce dependence on this assumption, by modeling inter-ROI relationships through graph connectivity rather than a fixed sequence. This is also a *neuroscientific* domain-specific question.

Use of Paired ROIs. A related assumption concerns the pairing of left and right hemisphere ROIs as input units for the BRNN and GNN. Whether such pairing is neuroscientifically justified is not straightforward. The Multibranch GNN was introduced as an attempt to address this, by processing each hemisphere independently before combining the representations.

Label Quality. The automatic label mapping procedure, based on WordNet hypernym hierarchies and sentence embeddings, introduced a degree of label noise. A manual annotation process, as adopted in [3], would have eliminated this source of error. Incorrect labels directly reduce classification performance, independent of the quality of the decoding models. This is primarily a *data* limitation, which could be addressed with research on machine learning models for this specific task. Since it was not the main topic of the project, but only a preliminary part, I decided for a simple solution (e.g. not considering COCO), and this resulted in a decrease in the number of training data. As a future development, it could be interesting to address also this task.

Dataset Size. The dataset may be too small relative to the complexity of the task. Only 4 subjects were available, and the number of trials per subject is limited. This likely contributes to the generalization problems observed across all models.

Future Direction: Voxel Selection. An interesting direction for future work would be to study how performance varies as a function of the number of selected voxels k in the top-k procedure. A larger k preserves more information but may introduce noise; a smaller k retains only the most discriminative voxels but risks discarding useful signal. A systematic comparison across values of k could clarify this trade-off.

Future Direction: Decoding Models. Due to limited available time, this work only proposed two architectures: BRNN and Multibranch GNN. While grid searches for hyperparameter tuning have been implemented, I did not spend much time in exploring different architectures, layers and modules. It could definitely be a future development of the work, which may lead to better results.

The project has been developed autonomously, without use of plagiarism.

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